

Tracking Memorization Geometry throughout the Diffusion Model Generative Process

Jonathan Brokman*

JONATHAN.BROKMAN@FUJITSU.COM

Itay Gershon*

ITAY.GERSHON@FUJITSU.COM

Omer Hofman

OMER.HOFMAN@FUJITSU.COM

Guy Gilboa

GUY.GILBOA@EE.TECHNION.AC.IL

Roman Vainshtein

ROMAN.VAINSHTEIN@FUJITSU.COM

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Abstract

Memorization in generative text-to-image diffusion models is a phenomenon where instead of valid image generations, the model outputs near-verbatim reproductions of training images. This poses privacy and copyright risks, and remains difficult to prevent without harming prompt fidelity. We present a mid-generation, geometry-informed criterion that detects, and then helps avoid (mitigate), memorized outputs. Our method analyzes the natural image distribution manifold as learnt by the diffusion model. We analyze a memorization criterion that has a local curvature interpretation. Thus we can track the generative process, and our criterion’s trajectory throughout it, to understand typical geometrical structures traversed throughout this process. This is harnessed as a geometry-aware indicator that distinguishes memorized from valid generations. Notably, our criterion uses only the direction of the normalized score field, unlike prior magnitude-based methods; combining direction and magnitude we improve mid-generation detection SOTA by $\sim 5\%$. Beyond detecting memorization, we use this indicator as a plug-in to a mitigation policy to steer trajectories away from memorized basins while preserving alignment to the text. Empirically, this demonstrates improved fidelity–memorization trade-off over the competitors. By linking memorization to magnitude-invariant geometric signatures of the generative process, our work opens a new direction for understanding—and systematically mitigating—failure modes in diffusion models.

1. Introduction

Understanding the geometry of natural-image distributions has been studied for almost a century [Simoncelli \(2024\)](#), where its roots lie in power-spectrum/statistical regularities and multiscale structure [Field \(1987\)](#); [Ruderman and Bialek \(1994\)](#); [van der Schaaf and van Hateren \(1996\)](#); [Huang and Mumford \(1999\)](#); [Portilla and Simoncelli \(2000\)](#); [Simoncelli and Olshausen \(2001\)](#); [Zoran and Weiss \(2012\)](#); [Wainwright and Simoncelli \(2000\)](#). In today’s age of generative AI, diffusion models enable sampling from this probability with unprecedented quality. They do so by approximating the probability’s score field, and simulating a multi-step reverse diffusion *generative process* [Hyvärinen \(2005\)](#); [Vincent \(2011\)](#); [Song and Ermon \(2019\)](#); [Ho et al. \(2020\)](#); [Song et al. \(2021\)](#); [Dhariwal and Nichol \(2021\)](#); [Rombach](#)

* Equal contribution.

et al. (2022). In accordance, a growing body of work develops theory and analyses of this approximated manifold Bortoli et al. (2022); Brokman et al. (2024b); Kadkhodaie et al. (2024); Benton et al. (2024); Sakamoto and Suzuki (2024); Tang et al. (2024a); Potapchik et al. (2024). Both high-quality sampling as well as score-function analysis of the learned natural image manifold are prominent features of *memorization* research.

Alongside the success of diffusion models came the prominent risk of *memorization*: near-verbatim reproduction of training images, which pose copyright and privacy risks Carlini et al. (2023); Webster (2023); Hu and Pang (2023); Tang et al. (2024b); Zhai et al. (2024); Pang et al. (2025). Recently, approaches analyzing the manifold geometry around points of memorization have shown significant potential in understanding and mitigating this phenomenon Wen et al. (2024); Jeon et al. (2025); Brokman et al. (2025). The former two emphasize the importance of geometry of memorization from *mid-generation* signals, so that it can be detected to steer away generation from memorized outcomes while sampling. Both offer methods that incorporate *magnitude* of score-functions.

This work advances the geometry-aware research of these phenomena. Our main contributions are as follows: ¹

- A novel **magnitude-invariant mid-generation detection criterion** based on the direction (not the size) of the classifier-free-guidance (CFG) guiding score field. This criterion, κ^Δ , approximates a high-dimensional generalization of the surface *mean curvature* - thus providing direct geometric interpretability.
- **Shifting-geometry observations.** Tracking κ^Δ over generation steps reveals a clear phase-structured behavior: memorized prompts start in an attracting region ($\kappa^\Delta > 0$) and grow rapidly toward a dominant mid-late concavity peak, whereas non-memorized prompts typically begin in a repelling convex regime ($\kappa^\Delta < 0$) before gradually crossing into a weaker concave regime that peaks around the same time. The separation is therefore governed by *early sign-geometry* and *later concavity strength*.
- **Improving the baseline.** Since our criterion holds information about the direction of $\nabla \log p(x)$, it holds complementary information to the previous magnitude-based approaches. Thus, it is natural to combine it with such methods - raising AUC at *earliest* generation steps from 0.925 to 0.97. Additionally - we demonstrate how plugging our criterion, as a stand-alone, into a mitigation strategy improves the fidelity-memorization balance over the competitors.

2. Related Work

Prior work has shown that diffusion memorization can be revealed through mid-generation signals. Wen et al. (2024) detect memorized prompts by measuring the magnitude of the classifier-free guidance (CFG) score gap throughout the denoising process, demonstrating that memorization produces abnormally large conditional-unconditional score norms even at the first generation step. Jeon et al. (2025) further analyze memorization via the

1. Official implementation: <https://github.com/JonathanBrok/Tracking-Memorization-Geometry-throughout-the-Diffusion-Model-Generative-Process/tree/main>

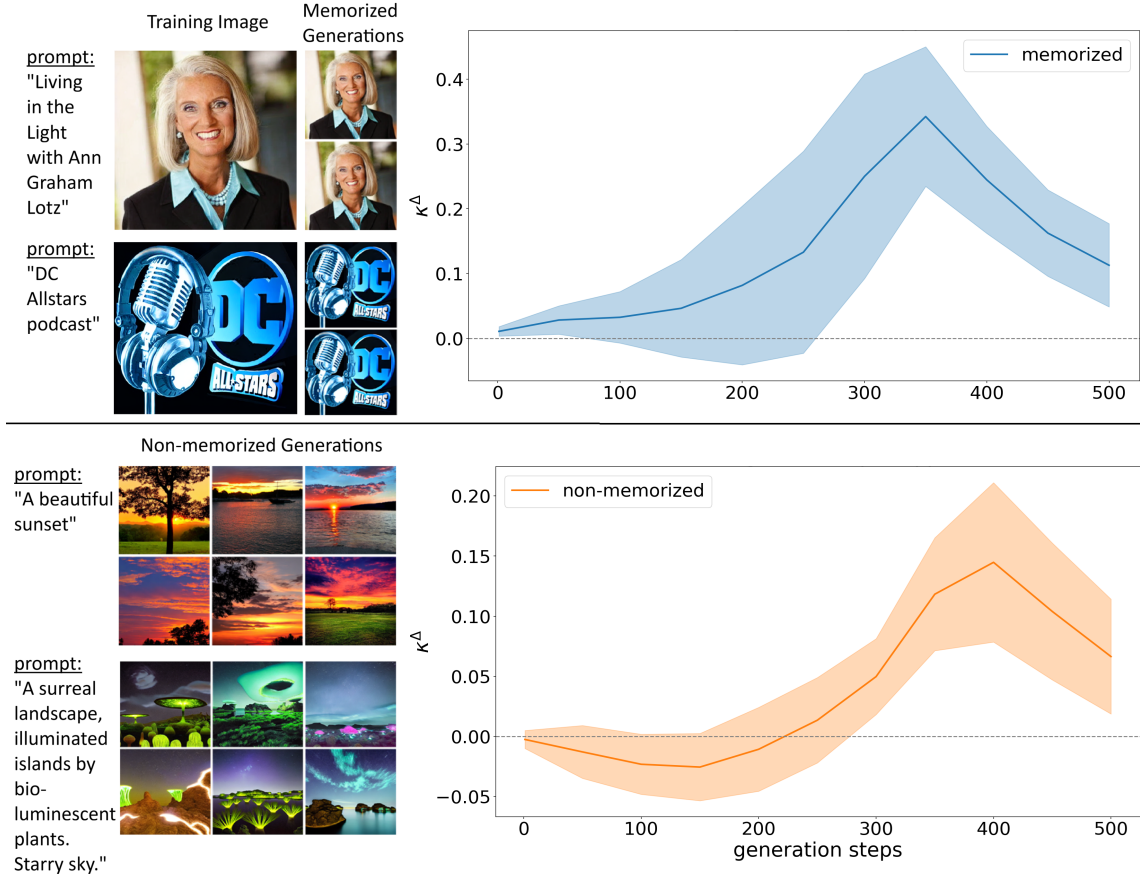


Figure 1: **Left:** Memorized prompts generate near-verbatim copies of a training images, unlike non-memorized prompts. **Right:** Temporal evolution of the curvature signal κ^Δ across generation steps. Memorized prompts immediately enter an attracting (concave) regime and steadily rise to a strong mid-late concavity peak. Non-memorized prompts typically begin in a repelling (convex) regime with $\kappa^\Delta < 0$, but later transition into a weaker positive concavity peak occurring in a similar time window. The distinction therefore lies in **early-phase geometry (sign)** and **mid-phase concavity strength**.

sharpness of the conditional probability landscape, using Hessian–score products. Both approaches rely on magnitude-dependent differentials of the score field. In contrast, our method introduces a magnitude-invariant, geometry-aware criterion based solely on the *direction* of the CFG difference. This yields a signed curvature surrogate, enabling us to track how prompt conditioning bends the generative trajectory’s local geometry and to combine this signal with magnitude-based methods for improved early-step detection and mitigation. Closest to our perspective, [Brokman et al. \(2025\)](#) propose p-Laplace-based detectors that apply higher-order differentials of the *final* score field, including the total-variation-based mean-curvature surrogate that we employ, in the spirit of classical image

and shape-processing methods [Sochen et al. \(1998\)](#); [Brokman and Gilboa \(2021\)](#); [Brokman et al. \(2024a\)](#). However, it is a post-hoc approach and is not suited for *mid-generation* steering (demonstrated in experiments below), or mitigation in general - since mitigation requires interfering during and not after the generative process. Other less related approaches intervene without modeling the geometry of the probability manifold: cross-attention analyses reveal token-level triggers and enable editing-time mitigation [Ren et al. \(2024\)](#); neuron-level methods identify and suppress memorizing units [Hintersdorf et al. \(2024\)](#).

In contrast to magnitude-dependent norms or post-hoc image tests, our geometry-aware criterion is both *magnitude-invariant* and *trajectory-aware*: we normalize the conditional-unconditional gap and track and approximate a well-known high-dimensional generalization of the surface mean curvature along the generative path. hence - tracking its signed value along the generative process has a direct geometrical interpretations.

3. Method

3.1. Diffusion model settings

Let $X \subset \mathbb{R}^d$ be data drawn from an unknown distribution with density p . Diffusion models learn the *score function* $s(x) := \nabla_x \log p(x)$ at noise level t , typically via noise prediction: Given a noising process $q(x_t | x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t)I)$, the MMSE noise predictor aligns with $-s_t(x_t)$. A neural network trained to predict noise thus estimates the score over time steps. See [Miyasawa et al. \(1961\)](#); [Nichol and Dhariwal \(2021\)](#).

Forward and reverse processes. The forward process progressively corrupts x_0 into noise, and its evolving densities q_t satisfy the Fokker–Planck equation associated with the chosen diffusion. The reverse process inverts this evolution: starting from $x_T \sim \mathcal{N}(0, I)$ ($\alpha_T = 0$), it uses the learned scores to reconstruct x_0 ($\alpha_0 = 1$) by following the reverse-time diffusion whose drift is defined by the Fokker–Planck time-reversal formula. See [Anderson \(1982\)](#); [Ho et al. \(2020\)](#); [Song et al. \(2021\)](#).

SDE view in text-conditional latent diffusion. In text-conditional settings, classifier-free guidance (CFG) provides unconditional and conditional scores, $s_\theta(x_t)$ and $s_\theta(x_t, c)$. The reverse dynamics can be written as an SDE driven by the score. In latent diffusion, the SDE process runs in a learned latent z_t , with decoding $x_0 = D(z_0)$. All quantities below apply identically in latent space by replacing x with z . See [Rombach et al. \(2022\)](#); [Ho and Salimans \(2022\)](#).

3.2. Method: Decode–encode boundary flux of classifier-free guidance

We measure how prompt conditioning deforms the log-probability landscape by estimating a mean-curvature analogue of $\log \frac{p(z_t|c)}{p(z_t)}$. This is done through a normalized boundary-flux divergence of the classifier-free guidance score gap in latent space.

CFG score gap. For a latent z_t and prompt c , let $s_\theta(z_t, c)$ and $s_\theta(z_t)$ be the conditional and unconditional scores. Their classifier-free guidance gap, and its normalized version are

$$s_t^\Delta(z_t, c) := s_\theta(z_t, c) - s_\theta(z_t), \quad \hat{s}_t^\Delta(z) := \frac{s_t^\Delta(z, c)}{\|s_t^\Delta(z, c)\|_\delta},$$

with $\|v\|_\delta = \sqrt{\delta + \|v\|^2}$ ensuring stability ($\delta > 0$ ensures a strictly positive denominator).

Boundary-flux curvature. Let $B_{R_t}(c_t)$ be a small latent ball centered at c_t with radius R_t , and let n be the outward unit normal on $\partial B_{R_t}(c_t)$. The curvature-like quantity we estimate is the normalized flux

$$\kappa^\Delta(c_t) := \frac{1}{|B_{R_t}|} \int_{\partial B_{R_t}(c_t)} \hat{s}_t^\Delta(z) \cdot n \, ds,$$

which limits to the mean curvature analogue $\nabla \cdot \hat{s}_t^\Delta(c_t)$ as $R_t \rightarrow 0$. Using N Monte Carlo samples $y_i \in \partial B_{R_t}(c_t)$ with normals $n_i = (y_i - c_t)/R_t$,

$$\hat{\kappa}^\Delta(c_t) \approx \frac{d}{R_t N} \sum_{i=1}^N \hat{s}_t^\Delta(y_i) \cdot n_i.$$

Scaled centers and decode–encode refinement. In latent diffusion, $z_t \approx \sqrt{\bar{\alpha}_t} z_0 + \text{noise}$, suggesting the geometry be probed at

$$c_t = \sqrt{\bar{\alpha}_t} z_0, \quad R_t = \sqrt{1 - \bar{\alpha}_t} R_0.$$

In [Brokman et al. \(2024b\)](#) they observed, in a different context, that processing SD latent space via a decode–encode process may improve separability. To retain this benefit in latent space, we optionally replace z_0 by a decode–encode refinement

$$\hat{z}_0 := E(D(z_t)), \quad c_t := \sqrt{\bar{\alpha}_t} \hat{z}_0. \quad (1)$$

The estimator applies equally to z_0 or $\hat{z}_0$, requiring only evaluations of $\hat{s}_t^\Delta$ in latent space.

Interpretable and practical curvature aspect. Since $s_\theta(z_t, c) \approx \nabla_{z_t} \log p(z_t | c)$ and $s_\theta(z_t) \approx \nabla_{z_t} \log p(z_t)$, our estimator satisfies

$$\kappa^\Delta(c_t) \approx \nabla_{z_t} \cdot \left(\frac{\nabla_{z_t} \log \left(\frac{p(z_t|c)}{p(z_t)} \right)}{\left\| \nabla_{z_t} \log \left(\frac{p(z_t|c)}{p(z_t)} \right) \right\|_\delta} \right),$$

i.e. it measures a **time-sensitive** mean-curvature analogue of the prompt-induced deformation of the latent probability geometry. Being t -sensitive, we can track $\kappa^\Delta(z_t)$ along the generation trajectory to profile how prompt-induced curvature evolves over time. Such trajectories were gathered in [Fig. 1](#).

For interpretability, we can discuss the observed dynamics ([Fig. 1](#)) as if we are traversing a 2D surface in 3D. Then $\kappa^\Delta = \nabla \cdot \hat{s}^\Delta$ acts as a signed mean-curvature. Let the principal curvatures be κ_1, κ_2 so

$$M := \frac{1}{2}(\kappa_1 + \kappa_2), \quad G := \kappa_1 \kappa_2.$$

are the *mean-curvature* (M), and *Gaussian curvature* (G). Usually, in simple 2-surface analysis (embedded in 3D) we perform full surface analysis by combining M and G to classify elliptic, hyperbolic, or parabolic points, but since κ^Δ captures only the mean-curvature

component it cannot provide such classification; nonetheless, its boundary-integral form exposes whether the prompt induces elliptic-like behavior (consistent inward/outward bending of normalized flux) or hyperbolic-like behavior (mixed bending directions), with the sign of κ^Δ indicating whether local behavior is more towards convex vs. concave bending of probability mass in latent space.

Numerical meaning. Numerically, κ^Δ aggregates how much the (normalized) score gap points *inward* to the local center - effectively summing the inward-facing component of the approximated probability gradient over a small spherical neighborhood. Positive values therefore indicate that conditional guidance bends probability mass inward (an attracting geometry), while negative values correspond to outward-bending or repelling behavior.

Memorized. κ^Δ is positive from the earliest steps, reflecting an attracting concave-leaning geometry. It rises steadily and forms a pronounced peak in mid-late generation, then decreases slightly while remaining positive.

Non-memorized. κ^Δ begins negative, corresponding to a repelling (convex-leaning) geometry. Over time it gradually increases, crosses zero, and enters a weaker concave-leaning regime with a smaller positive peak.

Bottom line. Memorized and non-memorized trajectories tend to have opposite signs in early steps, are *concave and convex-leaning respectively at the start*. Later non-memorized transitions into the attracting behaviour, but with weaker concavity compared to the memorized case, where both exhibit their strongest concavity in a similar time window.

4. Experiments

Let us evaluate the ability of κ^Δ and related metrics to detect memorized generations.

4.1. Settings and frameworks

We evaluate on *Stable Diffusion v1.4* with classifier-free guidance (CFG) scale 7.5. Unless stated otherwise, all detectors are computed at the first denoising step ($t=T$), which is the most actionable point for mitigation.

Prompt sets. Memorized prompts follow Wen et al. (2024); non-memorized prompts are LLM-generated to avoid overlap with training data Wen (2025).

Detection. Here we advocate for the decode–encode refinement (Eq. (1)). We compare our direction-only criterion κ^Δ against: (i) CFG-MAG—the score-norm gap under CFG Wen et al. (2024); (ii) HCP-MAG—the Hessian–CFG score product magnitude Jeon et al. (2025); and (iii) PLAP—a p-Laplace detector designed for post-hoc detection ($t = 0$, last generation step, unlike here) Brokman et al. (2025).

Hybrid detectors. To test complementarity, we learn a SVM over the 2D feature space $[\kappa^\Delta, \text{baseline}]$ on a small held-out split and report test AUC. We also compare against a hybrid of CFG-MAG and HCP-MAG.

Mitigation framework. Here we keep it simple, without encode–decode refinement. Following Wen et al. (2024), we perform soft-prompt optimization at $t=T$ for memorized prompts:

$$L(c) = L_{\text{miti}}(x_T, c) + \lambda \|c - c_0\|_2,$$

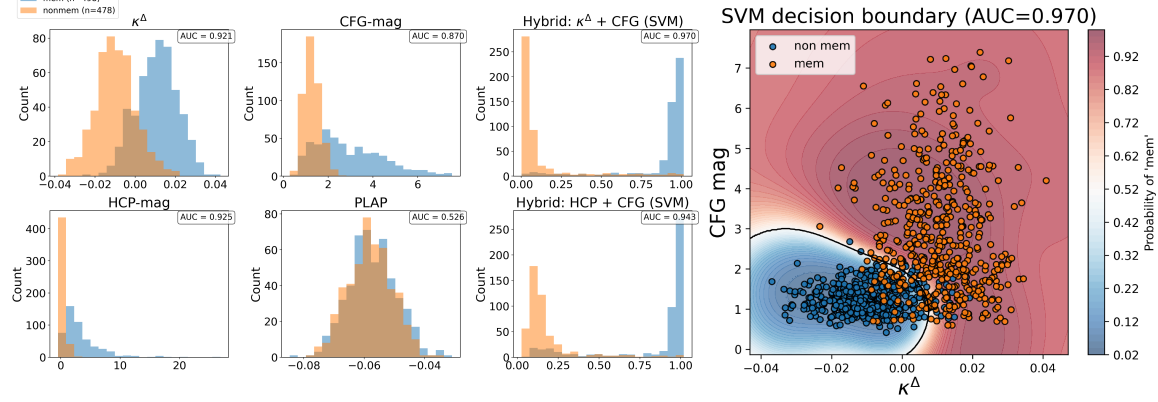


Figure 2: **Methods comparison - detection at $t=T$ (first generation step): Univariate and 2D hybrid.** Left: Histograms for memorized (orange) and non-memorized (blue) generations with AUCs overlaid. Our κ^Δ separates well (AUC = 0.922), CFG magnitude is competitive (AUC = 0.885), HCP magnitude is slightly higher in isolation (AUC = 0.932), while the post-hoc PLAP signal is uninformative at this early generation step (AUC = 0.550). Right: scatter of $x=\kappa^\Delta$ vs. $y=\text{CFG-mag}$ with a linear SVM and regions. The hybrid ($\kappa^\Delta + \text{CFG-mag}$) gives the best performance (AUC = 0.976), exceeding either component alone and a magnitude-only hybrid (HCP-mag + CFG-mag, AUC = 0.948). Take-away: *Directional* geometry is highly informative at the first step and complements magnitude-based criteria to achieve SOTA.

where c_0 is the initial text embedding and $x_T \sim \mathcal{N}(0, I)$. We instantiate L_{miti} with κ^Δ (ours) or each baseline criterion and compare fidelity–memorization trade-offs. Other generation hyperparameters follow the respective works.

4.2. Results

First generation step ($t=T$) separability. Figure 2 summarizes detection at the first denoising step, the most critical regime for on-the-fly mitigation. Our curvature surrogate κ^Δ already yields strong separability between memorized and non-memorized prompts (AUC = 0.922), outperforming the CFG magnitude baseline (CFG-mag, AUC = 0.885). The Hessian–score magnitude (HCP-mag) is slightly higher in isolation (AUC = 0.932), while the post-hoc 1-Laplace proxy (PLAP) is not informative at $t=T$ (AUC = 0.550), as expected for a criterion designed for post-generation.

Complementarity of direction and magnitude. Because κ^Δ depends on *direction* (normalized field) and CFG/HCP depend on *magnitude*, the signals are complementary. A simple linear SVM trained on a small held-out calibration split combining (κ^Δ , CFG-mag) achieves the best mid-generation discrimination (AUC = 0.976), improving over either component alone and also over the hybrid (HCP-mag + CFG-mag) (AUC = 0.948). The 2D scatter and decision regions in Figure 2 (right) visualize this complementarity.

Temporal signature at scale. The population trajectories in Figure 1 show a consistent *geometric* pattern across hundreds of prompts per class: for memorized prompts, κ^Δ rapidly departs from a narrow band near zero and climbs to a pronounced positive peak at mid–late steps, whereas for non-memorized prompts it symmetrically dips to a negative trough at roughly the same time indices. Importantly, the histograms in Figure 2 reveal that the two populations are already well separated even when κ^Δ is close to zero, so the subsequent zero-to-peak excursion sharpens—rather than creates—the distinction between memorized and non-memorized trajectories.

Mitigation outcomes. Figure 3 shows performance for the ”Mothers influence on her young hippo” memorized prompt, showing clear trade-offs: κ^Δ (**ours**): concentrates samples in the *high-fidelity / low-memorization* band. Qualitatively, outputs remain on-prompt (hippos) without reproducing the training instance. **SAIL sharpness** $\|H^\Delta s^\Delta\|^2$: reduces memorization but at a substantial cost in prompt fidelity; many samples drift off the target semantics. **Gap norm** $\|s^\Delta\|^2$: exhibits a similar trade-off, with a more prominent possible semantic drift. Overall, κ^Δ delivers the most favorable fidelity–memorization balance among the three, aligning with our hypothesis that a magnitude-invariant, geometry-aware signal is suited for mid-generation steering.

5. Summary

We introduced a magnitude–invariant, geometry–aware detector for diffusion memorization based on the *normalized boundary flux* of the CFG gap, denoted κ^Δ , with an optional decode–encode refinement step. The quantity is interpretable as a mean–curvature surrogate of the learned log–density along the generation trajectory, and it isolates the directional effect. Empirically, κ^Δ reveals a consistent *geometric signature*: memorized prompts start

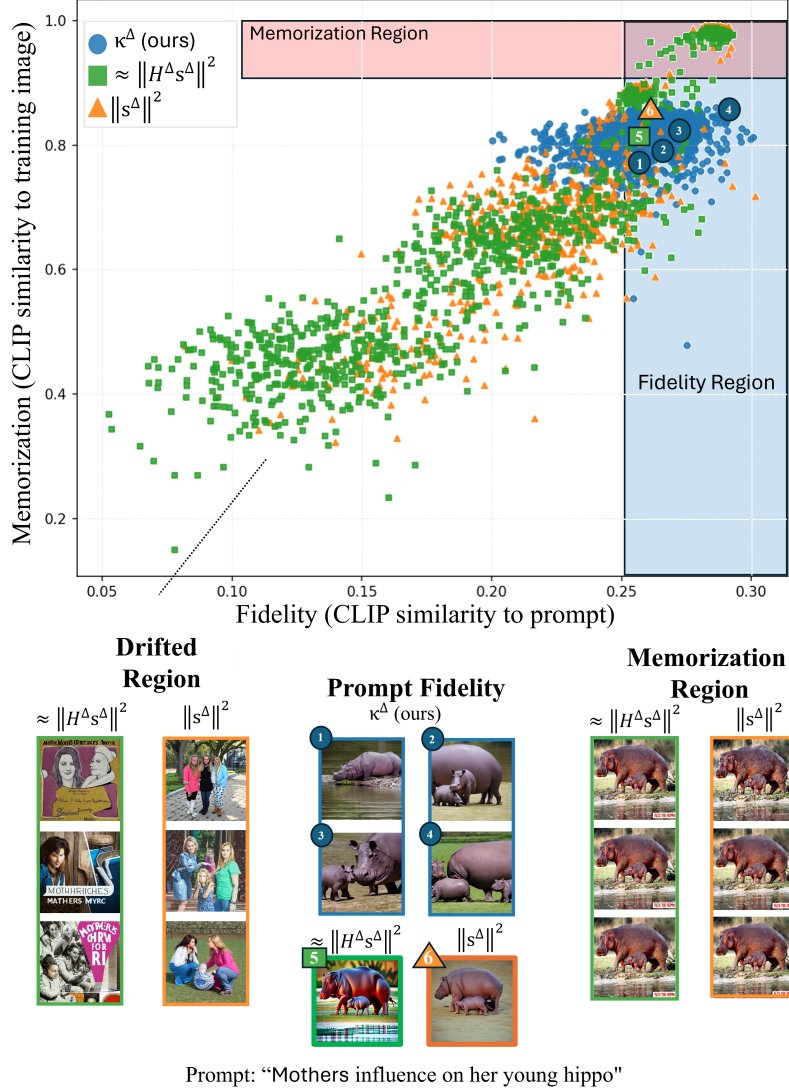


Figure 3: **Soft-prompt mitigation on a memorized prompt: A CLIP $\times$ CLIP of the fidelity–memorization trade-off**. Scatter shows *Fidelity* (CLIP cosine of generated image to the prompt) vs. *Memorization* (CLIP cosine of the generated image to the training image) for 3×1000 generations, plugging three criteria to Wen et al.’s soft-prompt optimizer: Our κ^Δ (blue), the SAIL criterion $\approx \|H^\Delta s^\Delta\|^2$ (green), and the gap norm $\|s^\Delta\|^2$ (orange). *Memorization Region*: The horizontal red band containing the top-right cluster - we verified that it contains all memorized images. *Fidelity Region*: the vertical band matching the fidelity of memorized copies. The desired mitigation outcome is in the blue rectangle (bottom-right): matching the prompt fidelity of the memorized image, with low similarity to the actual training image. Thumbnails below illustrate typical outcomes by region: *Drifted* (low fidelity), *Prompt-fidelity* (valid non-memorized generations), and *Memorization Region*. Numbered markers correspond to the highlighted examples. Compared with the competing criteria, our κ^Δ concentrates samples in the target region and reduces both drift and memorization.

in an attracting regime with $\kappa^\Delta > 0$ that strengthens until a mid-late peak, whereas non-memorized prompts begin in a repelling regime with $\kappa^\Delta < 0$ and only later transition into a weaker concave regime. Importantly, this entire behavior is expressed on the log-density ratio $\log\left(\frac{p(z_t|c)}{p(z_t)}\right)$. In terms of limitations: While κ^Δ is a novel mean-curvature proxy - if we want a complete analysis of surfaces, incorporating a generalization of the Gaussian curvature would be fitting; Additionally, we use the SD-1.4 model; broader validation across diverse architectures remains for future work. In terms of performance - we show across thousands of image generations that κ^Δ separates mem/non-mem already at the earliest step with strong AUC, and sets a new SOTA when combined with norm-based criteria such as $\|s^\Delta\|^2$. Turning the signal into a plug-in policy for soft-prompt optimization improves the fidelity-memorization trade-off relative to these baselines. Overall, this work proposes novel geometric signatures and advances memorization research of generative models - improving practical tasks with geometrical observations.

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